On the Evaluation of Structural Priors in Unrolled Channel Estimation for Rydberg Atomic Receivers

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***Abstract*—Adding an explicit structural prior to an unrolled channel estimator usually improves the reported metric, and the improvement is usually attributed to the prior. We show that in magnitude-only channel estimation for Rydberg atomic receivers, most of that improvement is an artifact of how such estimators are trained and scored. Normalized error averaged per sample over a mixed-SNR training set puts 89.7% of the gradient below 5 dB, leaving the high-SNR regime underfitted; a structural prior then supplies at high SNR what training did not. Train both arms adequately and a $+1.209$ dB structural advantage falls by a factor of fifteen, to $+0.078$ dB. The first casualty of this control was our own structured estimator. We propose a matched-adequacy protocol, show that simply rebalancing the loss recovers $+1.902$ dB at SNR ≥ 5 dB at no low-SNR cost, and show that pilot efficiency and pilot-count generalization are distinct quantities differing by 2.231 dB. Results are simulation only.**

***Index Terms*—Algorithm unrolling, channel estimation, evaluation methodology, phase retrieval, Rydberg atomic receiver, structural priors.**

I. Introduction

UNROLLING an iterative estimator into a trained network [1], [2] has become standard for channel estimation, including for Rydberg atomic receivers, where the magnitude-only readout makes the problem one of phase retrieval [3], [4]. A recurring next step is to add an explicit structural prior — a low-rank projection, a sparsity constraint — inside the unrolled loop, and to report the resulting improvement as evidence that the prior supplies information the network lacks.

This paper argues that such evidence is usually not decisive, and that the reason is a convention so widespread it is rarely stated: the training loss is a per-sample normalized error, averaged over a training set spanning a wide SNR range.

This is a finding about evaluation practice, and we did not arrive at it as critics. We built a Hankel-structured unrolled estimator, measured a clear $+1.209$ dB advantage at high SNR, and believed it. The control described in Sec. III was run to characterize that advantage, not to remove it. It removed it: the same contrast under matched training adequacy is $+0.078$ dB. The first casualty of the protocol we propose was our own method, and we report it because a result that survives an author's attempt to keep it is worth more than one that was never tested.

Contributions.

1) An attribution decomposition of an unrolled estimator's gain, separating the learned filter, the attention block, and unrolling itself against a matched non-unrolled control.
2) Identification of the confound: per-sample normalized error over mixed SNR concentrates the gradient at low SNR, and a structural prior compensates for the resulting high-SNR underfitting. We give a matched-adequacy protocol that separates the two.
3) A corrected loss that is not a workaround but a fix, recovering $+1.902$ dB at SNR ≥ 5 dB with no low-SNR penalty.
4) A demonstration that pilot efficiency and pilot-count generalization are distinct and are conflated by a single reported curve.

II. System Model and Estimator

An N-element uniform linear array serves K users over P pilots. With $\mathbf{G} \in \mathbb{C}^{N\times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the receiver observes magnitudes only,

$$\mathbf{Z} = \left|\mathbf{GS} + \mathbf{B} + \mathbf{W}\right|, \qquad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2). \tag{1}$$

Each user's channel is a sum of L_k plane waves on the array manifold, following the clustered Saleh–Valenzuela model [5]. The classical baseline is an expectation–maximization Gerchberg–Saxton iteration (EM-GS) [6]; the unrolled estimator ("URformer") replaces T iterations with T learned layers, each a data-consistency step followed by a learned filter, a gate, and a Transformer block across users [3].

We implement this estimator faithfully on the geometric ULA model of (1) rather than reproducing any published result: the operating point differs, and reaching the reported behaviour required roughly four times the stated data budget. Nothing here should be read as a reproduction or a validation of prior work, nor as a critique of any specific paper. The convention we examine is near-universal, and our own estimator uses it.

The reference curve is genie-aided. Where a reference appears, it is an unstructured least-squares estimate formed from the *true* phase of the noiseless field. It is not a bound: it presumes information no receiver has, and being restricted to unstructured least squares it can be passed by an estimator that exploits channel structure — which we in fact observe below 5 dB in Sec. IV-A.

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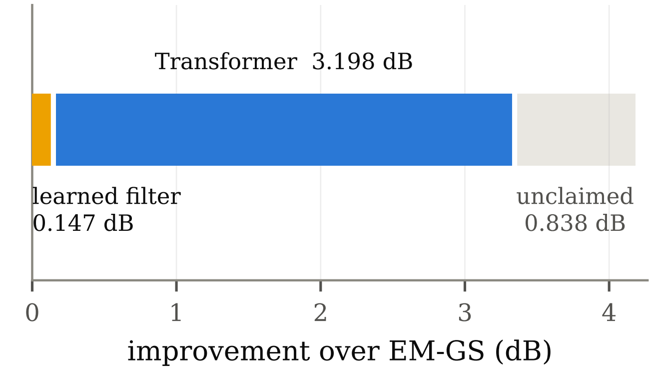

Fig. 1. Attribution of the unrolled estimator's gain over EM-GS at 80k training samples. "Unclaimed" is the remainder of the genie-aided least-squares headroom, which is a reference and not a bound.

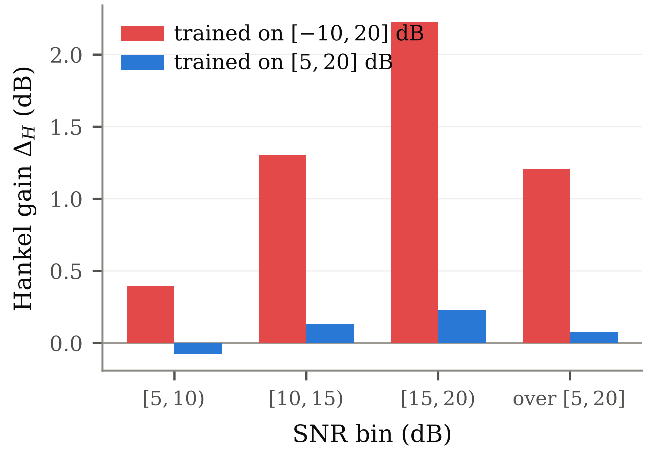

Fig. 2. The structural gain Δ_H under mixed-SNR training and under matched focused training. Both arms are retrained identically; only the training SNR range differs.

III. Attribution and the Matched-Adequacy Protocol

A. Where the gain comes from

Fig. 1 decomposes the unrolled estimator's 3.345 dB improvement over EM-GS. The learned per-layer filter, 980 parameters, contributes 0.147 dB; the attention block contributes the remaining 3.198 dB, or 96%. The filter's contribution does not grow with data — it falls slightly, $0.193 \to 0.147$ dB from 20k to 80k training samples — which is what a never-capacity-limited 980-parameter component should do.

Unrolling is doing real work independently of attention. Against a matched control that applies one Transformer block to a converged EM-GS estimate rather than unrolling, the unrolled estimator gains 0.920 dB, CI $[+0.888, +0.988]$, winning on 86.5% of trials.

B. The confound

The training loss is the per-sample normalized error $\|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 / \|\mathbf{G}\|_F^2$, averaged over a batch drawn across the whole SNR range. Because the normalized error of a low-SNR realization is orders of magnitude larger than that of a high-SNR one, this average is dominated by the low-SNR tail. Measured directly on our estimator, 89.7% of the gradient comes from trials below 5 dB, and the per-bin gradient share spans a factor of 31 across the range (Fig. 3, upper panel).

The consequence is that the high-SNR regime is systematically underfitted, and *any* mechanism that improves high-SNR behaviour will appear to add information. A structural prior is exactly such a mechanism.

C. The protocol

We propose the following control, which is cheap and which we recommend be run before any structural prior is credited.

1) Measure the per-bin gradient share of the training loss. If it is far from uniform, the regime receiving least gradient is underfitted.
2) Retrain *both* the structured and unstructured arms with training concentrated on the regime of interest, matching data budget, schedule, seed and initialization.
3) Re-measure the contrast. What survives is attributable to the prior; what does not was compensating for training inadequacy.

Applying it, and reporting what it did to our own method: over $[5, 20]$ dB the structural advantage under mixed-SNR training is $+1.209$ dB. Under matched focused training it is $+0.078$ dB, CI $[+0.050, +0.112]$ — a fifteenfold collapse (Fig. 2). The mechanism is visible in the absolute numbers: focused training gains the unstructured arm $+2.653$ dB but the structured arm only $+1.629$ dB. The unstructured network learns the array geometry by itself once it is given adequate gradient in that regime, leaving the explicit projection little to add.

Scope of this negative result. It holds at $N = 32$, $K = 3$, this data budget, and the two pilot counts tested. It is not a general claim that structural priors cannot help unrolled estimators. Indeed the same prior applied to the *classical* estimator, where no training adequacy is at stake, gives a large and stable gain, and under a distribution shift in path richness the structured network degrades least ($+0.48$ dB against $+0.73$ dB). The prior buys robustness here; it is the in-distribution accuracy claim that does not survive the control.

IV. Numerical Results

Table I lists the configuration. The reported statistic is the paired per-trial median within an SNR bin, with bootstrap 95% confidence intervals over 2000 paired test realizations.

A. A corrected loss, not a workaround

The confound is a property of the loss, so we fix the loss. We weight each sample by a static per-bin factor $w(b) = c/m(b)$, where $m(b)$ is the mean per-sample normalized error in bin b measured once before training, with c set so the weights have unit mean. This requires no schedule, no tuning, and no change to the architecture.

It works as designed. The realized per-bin gradient share spans a factor of 1.9 after balancing against 31 before, and

TABLE I
CONFIGURATION.

Parameter	Value
Array elements N / users K / pilots P	32 / 3 / 20
Paths per user L_k	$\mathcal{U}\{3,7\}$ [7]
SNR (train and test)	$\mathcal{U}[-10,20]$ dB
Reference-to-signal ratio	10 dB
Unrolled layers T	10
Trainable parameters	1,586,900
Training samples / epochs	80,000 / 13
Test realizations	2,000 (paired)

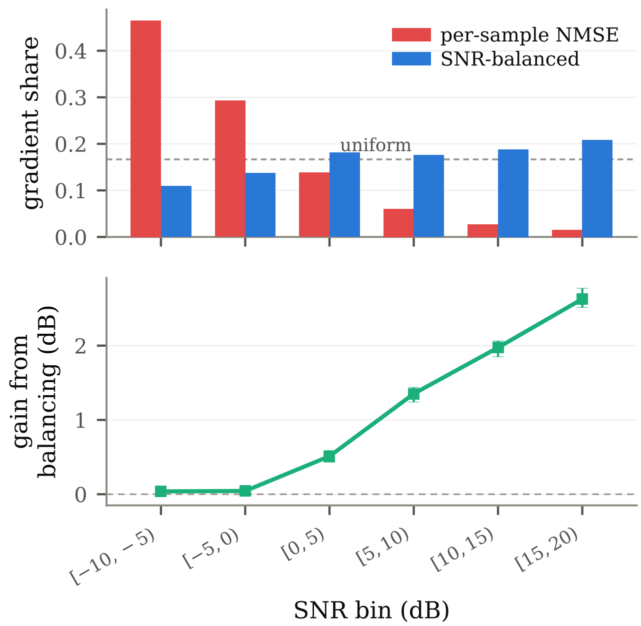

Fig. 3. Cause and effect. Upper: per-bin gradient share under the per-sample normalized loss and under SNR balancing. Lower: the resulting per-bin gain, paired per-trial median with bootstrap 95% CI.

the share below 5 dB falls from 0.897 to 0.427 against an ideal of 0.5 — a slight over-correction.

The estimation result is better than we predicted. Against an identically-trained unbalanced baseline, balancing gains $+1.902$ dB at SNR ≥ 5 dB, CI $[+1.823, +1.986]$, rising monotonically to $+2.628$ dB in the top bin. We had pre-registered a low-SNR *cost* of -0.10 to -0.40 dB on the reasoning that the lowest bin's weight falls about eighteenfold. There was no cost: the measured low-SNR change is $+0.135$ dB, CI $[+0.101, +0.172]$. Down-weighting the bins that dominated the gradient did not harm them, because they were over-served rather than merely well-served.

The gap to the genie-aided reference at SNR $\geq$ 5 dB falls from $+2.897$ to $+1.000$ dB. Below 5 dB the balanced estimator passes that reference by up to 1.99 dB, which is consistent rather than anomalous: the reference is restricted to unstructured least squares, and an estimator that has learned the channel geometry is not so restricted.

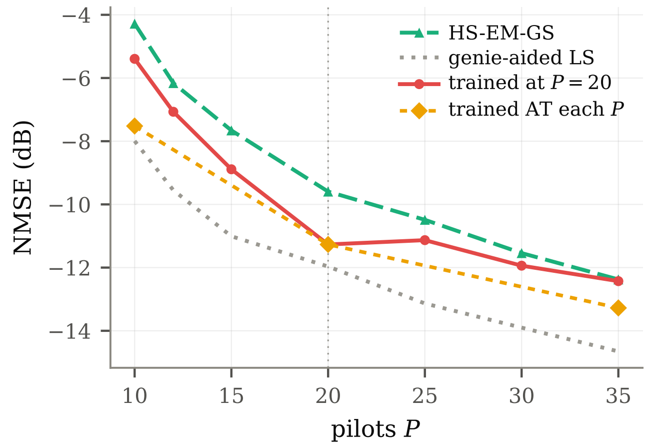

Fig. 4. Pilot efficiency against pilot-count generalization at 5 dB, on identical realizations. The dashed diamonds are separate networks trained at each P; the solid curve is one network trained at $P = 20$.

B. *Pilot efficiency is not pilot-count generalization*

A pilot-count curve for a trained estimator can mean two different things: one model trained at some P_0 and evaluated at each P (generalization), or a model trained at each P (efficiency). These are not close. Trained at $P = 10$, the estimator beats the $P = 20$ model evaluated there by $+2.231$ dB pooled, every bin's CI excluding zero. At $P = 35$ the corresponding margin is $+0.635$ dB, so the model generalizes upward in pilot count far better than downward.

At 5 dB the matched-trained estimator reaches -7.52 dB at $P = 10$, within 0.47 dB of the genie-aided reference, while the $P = 20$ model evaluated there is 2.60 dB short of it. The clearest single indication that the curve is tied to its training condition is that the $P = 20$ model is *worse* at $P = 25$ (-11.13 dB) than at $P = 20$ (-11.27 dB): more measurements, worse estimate. Fig. 4 shows all three curves together.

This matters for how such curves are reported. The pilot sweep of [3] is described as "the NMSE performance versus the number of pilot transmissions P, evaluated at a fixed SNR of 5 dB", and as "crucial for assessing the pilot efficiency of each algorithm"; the text does not state whether the networks were retrained per pilot count. We report that as a fact about what the paper specifies, and as motivation for stating the distinction explicitly — not as a criticism of that work, whose convention is the ordinary one and which our own earlier curves shared.

V. CONCLUSION

Most of the apparent benefit of adding an explicit structural prior to an unrolled channel estimator, in this setting, is an artifact of the training convention rather than information supplied by the prior. Per-sample normalized error over a mixed-SNR training set concentrates 89.7% of the gradient below 5 dB and leaves the high-SNR regime underfitted; a prior that helps there is credited with information it did not supply. Under a matched-adequacy control the effect falls fifteenfold, from

$+1.209$ to $+0.078$ dB. Rebalancing the loss — a change of a few lines, no architectural addition — recovers $+1.902$ dB at SNR ≥ 5 dB at no low-SNR cost, considerably more than the prior appeared to offer. We recommend the control be run before a structural prior is credited, and note that it cost us our own result first.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They cover the geometric ULA model of (1) at $N = 32$, $K = 3$, one data budget and the pilot counts tested; the negative result about structural priors is scoped to those conditions and is not a general claim. We establish no fundamental bound: the reference curve used throughout is genie-aided, presumes phase no receiver observes, and is passed by our own estimator at low SNR.

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